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Letter

CRISPR/Cas9-mediated mutagenesis of rice *NAC* transcription factor genes results in altered innate immunity

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Dear Editor,

Global climate change is beneficial to pathogen growth and adversely affects plant immunity, exacerbating the damage caused by diseases (Son and Park 2022). Bacterial blight caused by the bacterium *Xanthomonas oryzae* pv. *oryzae* (*Xoo*) and bakanae disease caused by the fungus *Fusarium fujikuroi* result in severe yield losses in rice (*Oryza sativa*). The plant-specific no apical meristem, Arabidopsis transcription activation factor, and cup-shaped cotyledon (NAC) transcription factor superfamily is involved in growth, development, and stress responses; it has more than 100 members present in many crops including rice (Maugarny-Cales et al. 2016; Singh et al. 2021). NACs are important for enhancing resistance in food crop plants via transgenic technologies because they have roles in growth and multi-environmental stress resistance (Singh et al. 2021). Therefore, *NAC* genes could serve as important targets for breeding plant varieties with enhanced resistance to various pathogens.

Multiplexing approach is an efficient strategy to create genetic diversity to counter climate change (Lorenzo et al. 2023). Samgwang is an elite Korean *japonica* rice cultivar bred by the Rural Development Administration and widely cultivated in South Korea. Thus, we attempted to develop mutant lines with edited *OsNAC* genes in the Samgwang background for plant improvement (Fig. 1A). We assembled clustered regularly interspaced short palindromic repeats (CRISPR)/CRISPR-associated nuclease 9 (Cas9) plasmids expressing single-guide RNAs for 146 *OsNACs* (Supplementary Table S1) and regenerated 1319 T_0 transgenic plants from hygromycin-resistant calli. The average genome-editing efficiency rate in CRISPR/Cas9-mediated T_0 transgenic plants was as high as 80.82% (Supplementary Fig. S1A), and 77.78% of the target genes showed a mutation rate of over 70% (Supplementary Fig. S1B). The rate of homozygous mutations was high, accounting for around a quarter of T_0 mutant plants (Fig. 1B). Elimination of the transgene occurred in 11.89% of T_1 homozygous rice mutant plants (Fig. 1C and Supplementary Fig. S2).

After selfing and molecular characterization of the CRISPR/Cas9-induced mutant lines, we obtained 60 T_1 homozygous mutant lines carrying mutations in the 23 *OsNAC* genes (Supplementary Table S2 and Fig. S3). To identify the *OsNAC* genes involved in innate immunity against bacterial blight, we inoculated the 60 T_1 homozygous mutants and wild-type plants with *Xoo* using a needleless syringe. The disease lesion length was reduced in *osnac30-1* compared to wild-type plants (Fig. 1D). Both *osnac30-1* and *osnac30-2*, which each contained a 1-bp insertion in the target site (Supplementary Fig. S3), showed enhanced resistance to *Xoo* compared to wild-type plants (Fig. 1E). To confirm this result, we analyzed the disease incidence of the *osnac30* mutant lines using the leaf-clipping

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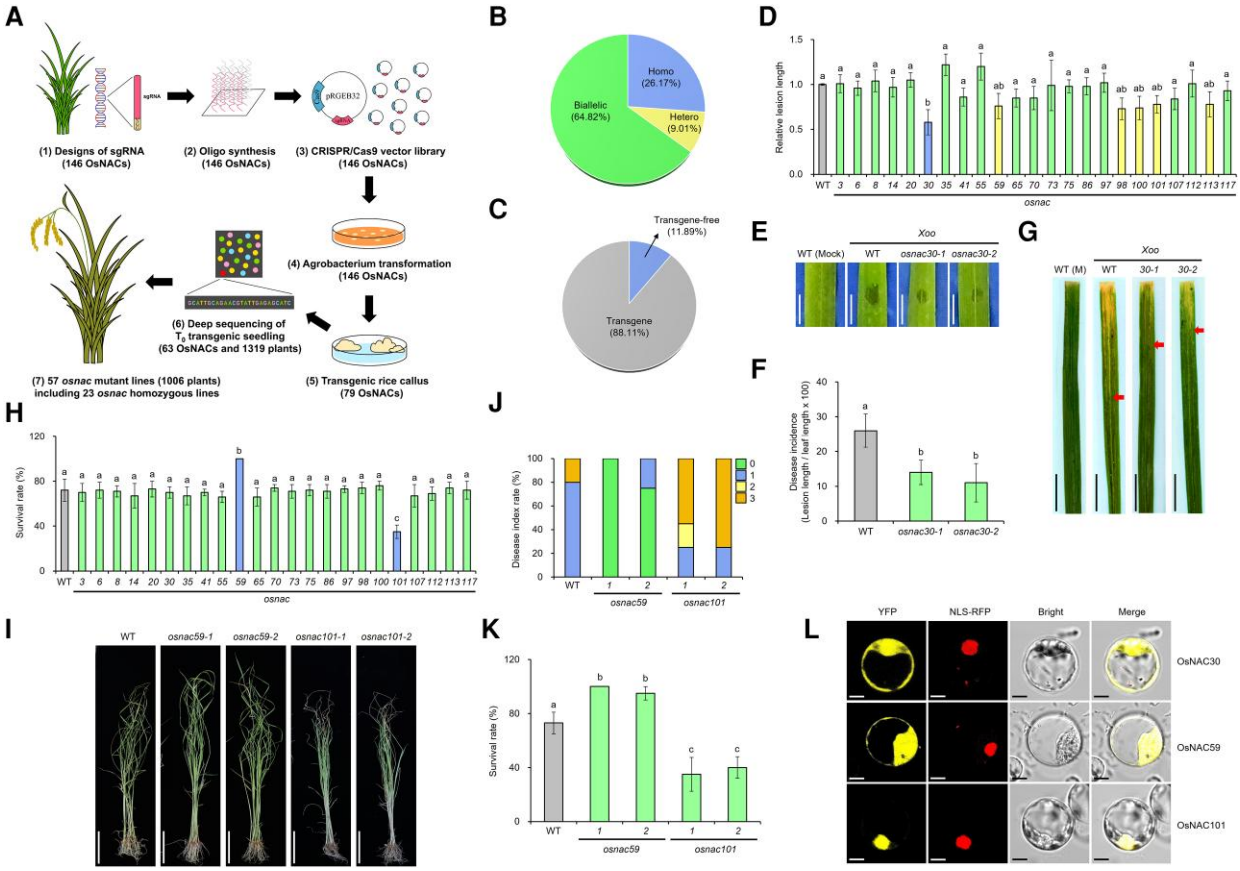


Figure 1. Development of CRISPR/Cas9-mediated *OsNAC* mutant lines with altered immunity. **A**) Schematic illustration of the method used to generate CRISPR/Cas9-mediated *OsNAC* gene-edited plants. **B)** Percentage of CRISPR/Cas9-mediated mutant genotypes within T_0 mutant plants. **C)** Percentage of transgene-free mutations within T_1 homozygous mutant plants. **D and E)** Phenotypic analysis of bacterial blight disease symptoms in the *osnac* mutant lines. Four-week-old T_1 homozygous *osnac* mutants and wild-type plants were inoculated with *Xanthomonas oryzae* pv. *oryzae* (*Xoo*) using a needleless syringe ($n = 8$). At 6 d postinoculation with *Xoo*, relative lesion length was calculated (**D**), and images of *osnac30-1* and *osnac30-2* were captured (**E**). Values are expressed as means ± SD. Different letters indicate significant differences according to ANOVA ($P < 0.05$). Scale bars, 1 cm. **F and G)** Eight-week-old *osnac30* mutants and wild-type plants were inoculated with *Xoo* using the leaf-clipping method ($n = 12$). At 14 d postinoculation, disease incidence was calculated (**F**) and images were captured (**G**). Values are expressed as means ± SD. Different letters indicate significant differences according to ANOVA ($P < 0.05$). Red arrow indicates the end of the lesion length. Scale bars, 2 cm. **H)** Phenotyping for bakanae disease symptoms in the *osnac* mutant lines. Six-day-old T_1 homozygous *osnac* mutant and wild-type plants were inoculated with *Fusarium fujikuroi* ($n = 10$). At 21 d postinoculation, survival rates were calculated. Values are expressed as means ± SD. Different letters indicate significant differences according to ANOVA ($P < 0.05$). **I to K)** Bakanae disease resistance of *osnac59* and *osnac101* mutant lines. The bakanae disease assay was carried out with 4-d-old T_1 homozygous mutants and wild-type plants ($n = 10$). At 21 d postinoculation with *F. fujikuroi*, images were captured (**I**), disease index was calculated (**J**), and survival rate was calculated (**K**). Disease index: 0, no symptoms; 1, blight symptoms on the leaf tip; 2, wilted shoots and partial leaf drying; 3, whole leaf was dry. Values are expressed as means ± SD. Different letters indicate significant differences according to ANOVA ($P < 0.05$). Scale bars, 2 cm. **L)** Subcellular localization of OsNAC30, OsNAC59, and OsNAC101. N-terminal yellow fluorescent protein-conjugated OsNAC30, OsNAC59, and OsNAC101 were introduced and expressed in rice protoplasts with nuclear localization signal-red fluorescent protein (NLS-RFP). After 12 h of incubation, images were captured under a confocal microscope. RFP signal indicates the nucleus.

method. The *osnac30* mutant lines showed markedly reduced susceptibility to bacterial blight compared to wild-type plants (Fig. 1, F and G). To identify *OsNAC* genes involved in innate immunity against bakanae disease, we monitored the phenotypes of homozygous T_1 mutant lines after *F. fujikuroi* inoculation. Mutation of *OsNAC59* conferred resistance to *F. fujikuroi*, while mutation of *OsNAC101* resulted in increasing susceptibility to this pathogen (Fig. 1H). *osnac59-1*, which harbored a 23-bp deletion in the target site, and *osnac59-2*, with a 1-bp deletion in the target site (Supplementary Fig. S3), showed greatly enhanced resistance to *F. fujikuroi* compared to wild-type plants (Fig. 1, I to K). However, *osnac101-1* and *osnac101-2*, each harboring a 1-bp insertion in the target site (Supplementary Fig. S3), showed significantly reduced innate immunity against *F. fujikuroi* compared to wild-type plants (Fig. 1, I to K). Moreover, OsNAC30 and OsNAC59 localized to the cytosol and nucleus, while OsNAC101 specifically localized to the nucleus (Fig. 1L). These results suggested NAC transcription factors can act as

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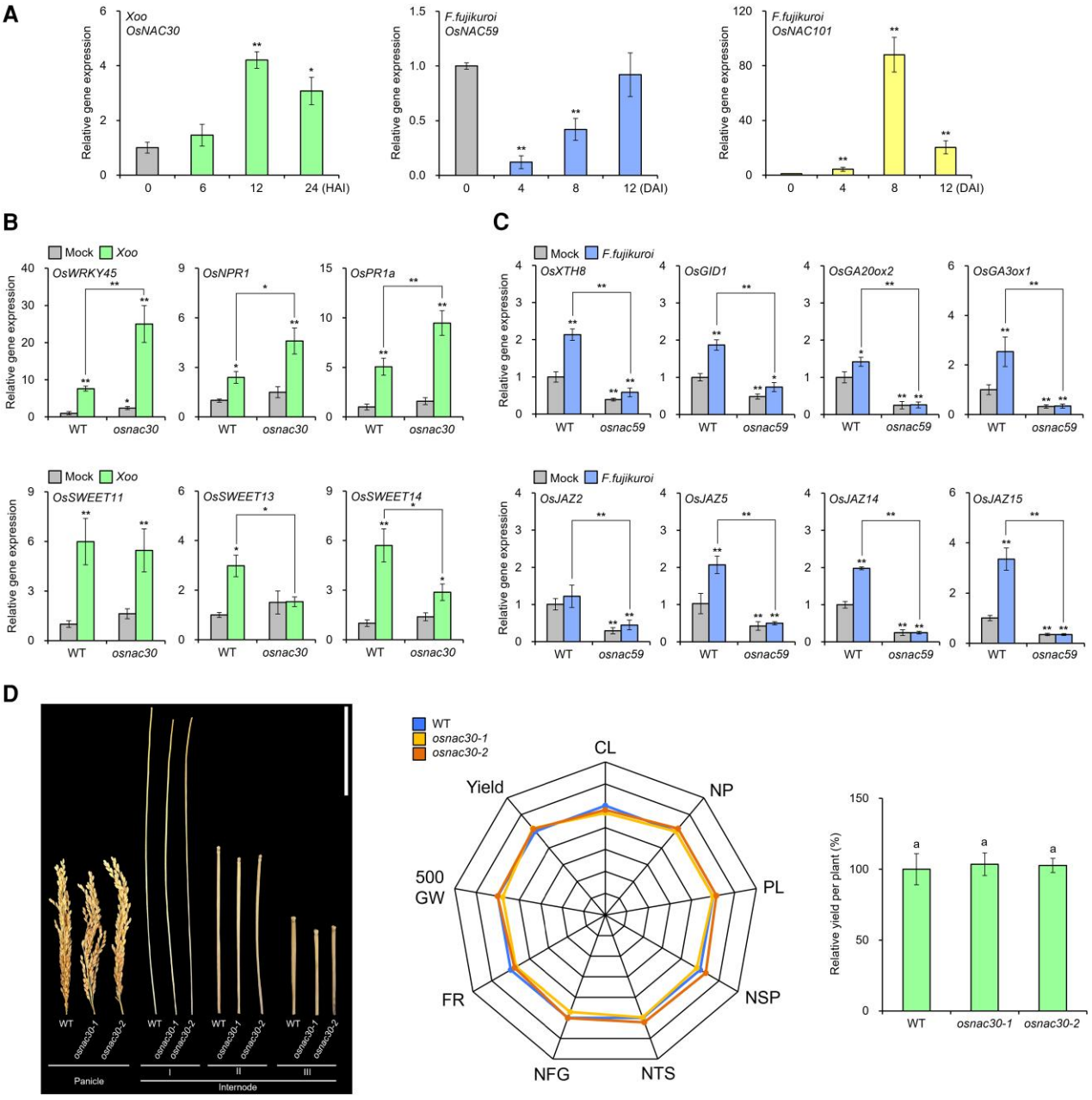


Figure 2. Gene expression analysis and agronomic traits of the *osnac* mutants. **A)** Relative expression levels of the *OsNAC* genes after *Xanthomonas oryzae* pv. *oryzae* (*Xoo*) or *Fusarium fujikuroi* inoculation. The wild-type plants were inoculated with *Xoo* ($n = 12$) or *F. fujikuroi* ($n = 10$), respectively, and collected after the indicated incubation times. Relative transcript levels of the *OsNAC* genes were determined by RT-qPCR. *OsActin* was used as an internal control. Values are expressed as means ± SD. Asterisks indicate values significantly different from that at 0 h according to *t*-test (*$P <$ 0.05, **$P < 0.01$). **B)** Expression levels of defense- and susceptibility-related genes in *osnac30*. Eight-week-old *osnac30* and wild-type plants were inoculated with *Xoo* ($n = 12$), and total RNAs were extracted from the leaves at 12 h postinoculation. Relative transcript levels of the genes were determined by RT-qPCR. *OsActin* was used as an internal control. Values are expressed as means ± SD. Asterisks indicate values significantly different from mock-treated wild-type samples and between the indicated samples according to *t*-test (*$P < 0.05$, **$P < 0.01$). **C)** Expression levels of GA- and JA-related genes in *osnac59*. Six-day-old *osnac59* and wild-type plants were inoculated with *F. fujikuroi* ($n = 10$), and total RNAs were extracted from the shoot tissues at 8 d postinoculation. Relative transcript levels of the genes were determined by RT-qPCR. *OsActin* was used as an internal control. Values are expressed as means ± SD. Asterisks indicate values significantly different from mock-treated wild-type samples and between the indicated samples according to *t*-test (*$P < 0.05$, **$P < 0.01$). **D)** Agronomic traits of *osnac30*. The plants were grown in a paddy field for 120 d, and then relative yield components such as culm length (CL), number of panicles per plant (NP), panicle length (PL), number of spikelets per panicle (NSP), number of total spikelets (NTS), number of filled grains (NFG), filling rate (FR), 500-grain weight (500 GW), and yield per plant (Yield) were measured ($n = 8$). Values are expressed as means ± SD. Significant differences between the indicated samples are calculated according to ANOVA ($P < 0.05$). Images were digitally extracted for comparison. Scale bars, 10 cm.

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negative or positive regulators of plant immunity via controlling the expression of genes involved in specific plant–pathogen interaction.

The expression level of *OsNAC30* was significantly induced 12 h after inoculation with *Xoo* (Fig. 2A). After *F. fujikuroi* inoculation, *OsNAC59* transcript level was dramatically decreased at 4 d after inoculation (DAI) (Fig. 2A). However, the expression of *OsNAC101* began to increase at 4 DAI and increased dramatically (almost 80-fold) at 8 DAI with *F. fujikuroi* (Fig. 2A). To explore why *osnac30* and *osnac59* show enhanced disease resistance, we performed gene expression analysis after *Xoo* and *F. fujikuroi* inoculation, respectively. The salicylic acid (SA)-mediated defense response is required for the immunity of rice against *Xoo* (Yang et al. 2013). The expression of susceptibility genes such as *SUGAR WILL EVENTUALLY BE EXPORTED TRANSPORTERS 11* (*OsSWEET11*), *OsSWEET13*, and *OsSWEET14* also plays a key role in the response of rice to *Xoo* infection (Oliva et al. 2019). Transcript levels of SA-dependent immunity marker genes such as *WRKY DNA-BINDING PROTEIN 45* (*OsWRKY45*), *NONEXPRESSOR OF PATHOGENESIS-RELATED GENES 1* (*OsNPR1*), and *PATHOGENESIS-RELATED 1A* (*OsPR1a*) were significantly higher in *osnac30* compared to wild-type plants after *Xoo* inoculation (Fig. 2B). *Xoo*-induced expression of *OsSWEET13* and *OsSWEET14* was markedly inhibited in *osnac30-1* plants compared to wild-type plants, while the transcript level of *OsSWEET11* in *osnac30-1* was similar to that of the wild-type (Fig. 2B). Plant hormones such as gibberellin (GA) and jasmonic acid (JA) play important roles in the rice–*F. fujikuroi* interaction (Matic et al. 2016; Cheng et al. 2020). *F. fujikuroi* increases GA levels in rice, leading to bakanae disease ("foolish seedling") symptoms, while increased JA-mediated defense responses contribute to resistance to *F. fujikuroi*. To examine GA and JA responses in *osnac59*, we inoculated *osnac59*-1 with *F. fujikuroi* and measured the expression levels of GA- and JA-related genes. The GA response marker gene *XYLOGLUCAN ENDOTRANSGLUCOSYLASE/HYDROLASE 8* (*OsXTH8*), which is specifically upregulated by GA (Jan and Komatsu 2006); the GA receptor gene *GIBBERELLIN INSENSITIVE DWARF 1* (*OsGID1*); and GA biosynthetic genes such as *GIBBERELLIN 20-OXIDASE 2* (*OsGA20ox2*) and *OsGA3ox1* were downregulated in *osnac59* compared to wild-type plants (Fig. 2C). *JASMONATE-ZIM-DOMAIN PROTEIN* (*JAZ*) genes (i.e. *OsJAZ2*, *OsJAZ5*, *OsJAZ14*, and *OsJAZ15*), encoding negative regulators of JA signaling, were also downregulated in *osnac59* compared to wild-type plants (Fig. 2C).

Enhanced Disease resistance often results in yield loss (Ning et al. 2017). Interestingly, the yield components of *osnac30* were similar to those of wild-type plant (Fig. 2D). This study provides the vital basis for analyzing the roles of OsNACs as well as valuable materials for breeding elite rice cultivar with enhanced innate immunity.

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Author contributions

S.S. and S.R.P. conceptualized and supervised the study. S.S., G.S., S.N., G.L., and J.I. performed the experiments and data analysis. S.S., G.S., S.N., G.L., J.I., K.S.L., Y.J.P., and E.-J.S. generated and verified the mutant plants. S.S. wrote the manuscript. All authors read and approved the manuscript.

Supplementary data

The following materials are available in the online version of this article.

Supplementary Figure S1. Genome-editing efficiency of CRISPR/Cas9-mediated T_0 transgenic plants.

Supplementary Figure S2. Isolation of transgene-free mutant lines.

Supplementary Figure S3. Mutations in the T_1 homozygous mutant lines.

Supplementary Methods S1. Methods used in this paper.

Supplementary Table S1. Sequences of single-guide RNAs and molecular characterization of *osnac* plants.

Supplementary Table S2. T_1 homozygous *osnac* mutant lines.

Supplementary Table S3. Sequences of the primers used in this study.

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Conflict of interest statement. None declared.

Data availability

The data supporting the findings of this study have been provided in the text and in the Supplementary data files and are available upon request.

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